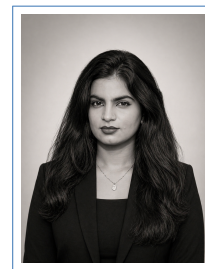


PRATISHTHA ABROL

Curriculum Vitae

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Education

2023-2025 : **Masters by Research, Quantum Science & Technology**, *International Institute of Information Technology*, Hyderabad, (IIITH).

Quantum Field Theory, Quantum Algorithms, Open Quantum Systems and Thermodynamics, Technical and Research Writing

Teaching Assistantship: Quantum Information and Computation, Entropy and Information

2019–2023 : **Bachelor of Technology, Computer Science**, *International Institute of Information Technology*, Hyderabad, (IIITH).

Algorithm Design and Analysis, Data and Applications, Operating Systems, Networking, Automata Theory, Game Theory, Machine Learning, Distributed Systems, Software Engineering and Foundations, Data Analytics, Modern Complexity Theory

Teaching Assistantship: Advanced Algorithms, Data and Applications,

Publications

Journal Articles

2025 Pratishta Abrol, Pahulpreet Singh, and Indranil Chakrabarty. Maximal secret reconstruction, teleportation and bell's inequality. *The European Physical Journal D*, volume 79, page 11. Springer, 2025.

Research Experience

Center for Quantum Science and Technology (CQST), IIIT-Hyderabad

June, 2021 – **Maximal Secret Reconstruction, Teleportation and Bell's Inequality.**

March, 2025 Identifying the best candidate among the secret sharing resource states in achieving the maximum reconstruction fidelity thus by setting the practical information transfer limit in a possible resource theoretic extension of secret sharing.

Advisor : **Dr. Indranil Chakrabarty**, Associate Professor, Center for Quantum Science and Technology (CQST), Center for Security, Theory and Algorithms (CSTAR), IIIT Hyderabad ([Faculty Webpage](#))

Software Engineering Research Center (SERC), IIIT-Hyderabad

July, 2021 – **Developing abstract use cases for enhanced Python algorithmic and data model understanding.**

July 2022 Analyzing and designing conceptual ideas behind Python data structures, concepts and algorithms. Visual representation of said ideas via computer graphics and Java Script programming.

Advisor : **Dr. Karthik Vaidyanathan**, Assistant Professor, Software Engineering Research Center (SERC), IIIT Hyderabad ([Personal Web-page](#))

Computer skills

Languages Python, R, C, C++, Advanced JAVA, Qiskit, Cirq, Bash

Web HTML 5, PHP, Javascript

ML/Data TensorFlow, SQL, MongoDB